

Implementation Identity Matters: Matched-Seed Post-Route Evidence for Memory ECC Hardware

Anonymous submission

Abstract—An error-correcting-code (ECC) label identifies coding semantics, not a unique hardware or physical realization. We present an identity-preserving evaluation method that qualifies correctness before physical measurement and applies one pinned RTL-to-route and equal-work power policy over 5 predeclared matched seeds. For exact-equivalent combinational and pipelined SECEDED (72,64), the pipeline changes area by 37.2% on average (range 36.7–37.8%), slack-derived frequency by 46.6% (42.1–55.3%), and steady-stream energy/op by -23.4% (from -24.0% to -22.5%). Area, frequency, and energy orderings occur in 5/5, 5/5, and 5/5 matched seeds, respectively; request latency remains one versus three cycles. We also prove exact behavioral identity for an algorithmic Hsiao decoder before routing it under the same sweep; it meets the 10 ns target in 5/5 runs. The evaluated shortened BCH (78,64, $t = 2$) realization meets that target in 0/5 runs. These are controlled heuristic-sensitivity envelopes, not population estimates or family-wide rankings. The results show that correctness evidence, RTL identity, target feasibility, and physical variability must remain explicit when comparing ECC hardware.

Index Terms—memory ECC, SECEDED, Hsiao, BCH, physical design, seed sensitivity, energy

I. INTRODUCTION

Memory ECCs are usually compared by code rate and guaranteed correction or detection capability. These properties are indispensable, but they do not fix encoder/decoder logic, register placement, latency, clocking, routing, or activity. Consequently, a code name is not a sufficient experimental unit: two designs can implement exactly the same input/output relation and occupy different physical trade-off regions, while one RTL instance cannot safely inherit correctness or PPA from another merely because their labels match.

Hardware-aware SECEDED construction and checker optimization are established [1]–[3]; stronger-correction codec cost and BCH microarchitecture are likewise established [4]–[7]. Cross-layer memory studies already trade reliability against device, array, storage, performance, or energy cost [8]–[11]. Our question is narrower: when correctness-qualified RTL instances share a transaction contract and a pinned post-route policy, do architecture orderings and effect magnitudes survive placement/routing heuristic variation?

We treat implementation identity as the experimental unit and predeclare five matched OpenROAD seeds. The central relation is

**ECC code identity $\neq$ hardware architecture identity
 $\neq$ physical implementation outcome.**

The exact-equivalent SECEDED pair isolates temporal microarchitecture. A newly algorithmic Hsiao decoder recovers a physical point only after exact equivalence to the historical qualified

behavior is proved. A correctness-qualified BCH realization supplies a stronger-guarantee, implementation-scoped feasibility contrast. Missing and timing-infeasible measurements remain explicit.

This paper contributes:

- 1) an auditable qualification and characterization method that prevents evidence from being pooled across code, RTL, and physical identities;
- 2) a prospective matched-seed experiment for an exact-equivalent SECEDED pair, including route, timing, and equal-useful-work energy sensitivity; and
- 3) a common-policy comparison with a same-code algorithmic Hsiao identity and one stronger-correcting BCH realization, with correction strength kept categorical.

The work does not propose a new ECC, claim that ECC/PPA co-evaluation is new, or estimate a universal distribution of routing outcomes.

II. RELATED WORK

Hsiao’s odd-weight-column construction makes checker complexity a code-design concern [1]. Ghosh *et al.* optimized Hamming/Hsiao checkers for power while monitoring area and delay [2], and Reviriego *et al.* constructed low-delay SEC/SECEDED logic [3]. Recent ASIC synthesis compares SECEDED and SECEDED-DAEC designs over several lengths [12]. These works establish hardware-conscious SECEDED; our contribution is not a lower-cost code, but a matched post-route test of architecture identity after semantic equivalence.

Naseer and Draper compared parallel SEC, Hsiao SECEDED, DEC, and DEC-TED hardware [4]; Layered-ECC targets low-complexity DEC decoding [6]. Wong *et al.* altered Chien-search power management [5], while recent direct and conventional BCH decoders show that architecture can materially change area efficiency and latency [7]. This prior art requires our BCH conclusion to remain tied to the evaluated syndrome/Chien RTL.

STT-MRAM studies co-design correction strength with device/array cost [8], [9]; LAD-ECC and Stealth ECC adapt protection using workload or data properties [10], [11]; FaultSim supplies field-model-oriented system reliability evaluation [13]. We instead hold protection semantics and useful work fixed, measure codec logic in a pinned open flow [14], and make no FIT, SER, array, or operational-failure claim. A focused 2018–2026 audit found no close paper combining exact RTL identity transfer, common post-route constraints, predeclared matched seeds, feasibility-gated energy, and explicit unavailable points.

We use that only to position the experiment; it is not a priority or universal-absence claim.

III. IDENTITY-PRESERVING METHOD

A. Qualification and Reliability Semantics

Figure 1 separates three identities. A *code identity* fixes the payload/codeword mapping and decoder semantics. A *hardware identity* fixes RTL, interface, latency, and initiation interval (II). A *physical identity* adds constraints, tools, collateral, and seed. Correctness evidence admits a hardware identity; favorable PPA cannot rescue an unqualified design.

Every admitted cross-layer claim is an explicit evidence join. The code record supplies the guarantee and exact payload/codeword mapping; the RTL freeze supplies source hashes, interface, latency, and II; the physical record supplies the run seed, configuration hashes, stage status, and raw reports. A join is valid only when these identities agree. This prevents three common substitutions: transferring a proof between nominally similar RTLs, presenting a routed result beside power from another netlist, or treating a failed target-clock analysis as an achievable energy point. The Hsiao recovery deliberately exercises this rule: semantic equivalence transfers, but physical measurements do not transfer between table and algorithmic hardware representations.

The two conventional SECDDED candidates implement the same systematic extended-Hamming (72,64) relation. Formal equivalence after a two-cycle temporal alignment proves identical encoder/decoder responses. Both accept independent encoder and decoder transactions at II=1; request latency is one cycle for combinational and three for pipelined RTL.

Historical Hsiao semantics used a qualified (72,64) parity-check identity, but its table representation inferred a 256×73 memory exceeding the unchanged 4096-bit synthesis-memory policy. Revision 2 reconstructs the exact column order and implements syndrome generation plus 72 explicit syndrome-to-column comparisons. Before PPA, a miter proves the algorithmic decoder and qualified reference produce identical outputs for every arbitrary 72-bit received word. Thus code-level guarantees transfer, while PPA belongs specifically to the new algorithmic hardware identity.

Formal equivalence is stronger here than enumerating only errors inside a declared guarantee. The Hsiao miter accepts an unconstrained 72-bit received vector, computes both decoders, and asserts equality of payload and status outputs. It therefore covers no-error words, codewords with any injected mask, and vectors that are not reachable from the selected encoder. A separate structural check confirms 72 equality comparators and no inferred memory, while the parity-check column sequence is byte-frozen. The proof completed and the RTL hash was recorded before any Hsiao physical run.

Conventional and Hsiao SECDDED correct every weight-1 coordinate error and detect every weight-2 error. The shortened BCH (78,64, $t = 2$) realization corrects every weight-1 and weight-2 error in its qualified universe. Correction strength remains categorical; it is never converted into an arbitrary scalar score. Exhaustive weight-3 observations remain artifact-level

decoder characterization because no physical fault weighting makes them operational rates.

B. Timing, Power, and Decision Rules

Authoritative ORFS reporting derives its clock metric from OpenSTA's minimum-period analysis with port paths included. We retain both that raw value and the signed worst setup slack. For the fixed implementation optimized under target $T = 10$ ns, we report

$$f_{\text{slack}} = \frac{1000}{T - s_{\text{setup}}} \text{ MHz}, \quad (1)$$

where s_{setup} is signed, not conventional WNS clipped at zero. This is a *post-route slack-derived frequency estimate*; it is not a frequency sweep, an experimentally searched Fmax, or a design re-optimized at the derived period. Target feasibility separately requires nonnegative setup slack and zero setup violations.

Power uses post-route OpenSTA with the same deterministic no-error VCD for each seed. Each timing-feasible SECDDED and Hsiao point receives 100000 useful II=1 operations, identical reset/drain semantics, the 10 ns clock, and common input activity. Energy is $E_{\text{op}} = PT_{\text{trace}}/N_{\text{useful}}$. BCH power at an infeasible target is diagnostic only; its target-clock energy is unavailable.

Within the SECDDED guarantee tier, dominance minimizes area, request latency, and achievable energy/op while maximizing f_{slack} . We enumerate relations independently per seed, at aggregate means, and over sensitivity ranges. No p -values or distributional interpretation are attached to five deterministic heuristic seeds.

The matched design controls comparison noise without claiming randomized inference. For each seed s , both SECDDED implementations receive identical values for global placement, global routing, and general OpenROAD randomness. Paired effects are computed before aggregation as $(x_{\text{pipe},s} - x_{\text{comb},s})/x_{\text{comb},s}$. We report all five values plus mean, median, sample standard deviation, extrema, and range. Pairing exposes whether an ordering flips under a common heuristic perturbation; the standard deviation is descriptive only. A predeclared poor seed remains in the envelope, and an infrastructure replacement is permitted only with the same seed and a preserved failed attempt.

Dominance is assessed three ways. Per-seed enumeration tests whether either implementation is no worse in every admitted objective at that realization. Aggregate-mean enumeration asks the same question of the two mean points. Finally, a conservative range check refuses to claim dominance unless sensitivity intervals are no worse in every objective. These checks cannot establish a universal Pareto frontier, but they prevent averaging from hiding seed-dependent reversals.

C. Prospective Controls and Missingness

The campaign matrix, seed rule, qualification gates, timing target, power eligibility, and analysis code were fixed before the first measured route. A completed but timing-infeasible route remains a physical observation: its area, routing, DRC, hold, signed setup slack, and setup-violation count are retained.

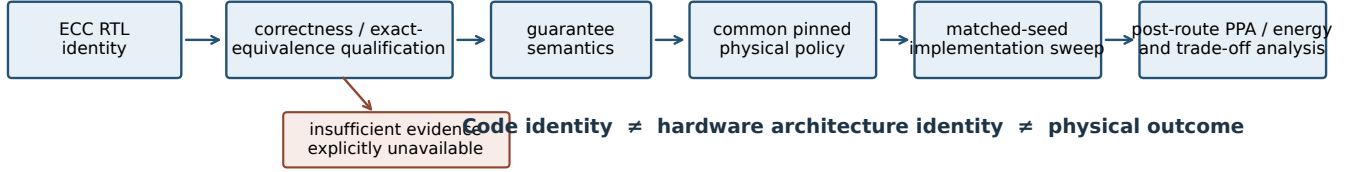


Fig. 1. Qualification precedes physical characterization. A failed evidence join remains unavailable; accepted identities enter one common matched-seed policy.

TABLE I
QUALIFIED IMPLEMENTATION IDENTITIES

Architecture	Bits	Guarantee	Lat.	II	Evidence
SECDED comb.	72	W1 correct; W2 detect	1	1	qualified; routed 5/5; 10 ns 5/5
SECDED pipe.	72	W1 correct; W2 detect	3	1	qualified; routed 5/5; 10 ns 5/5
Hsiao algorithmic	72	W1 correct; W2 detect	1	1	exact identity; routed 5/5; 10 ns 5/5
BCH (78,64,t=2)	78	W1/W2 correct	1	1	qualified; routed 5/5; 10 ns 0/5

It does not, however, become a target-feasible design through a derived-frequency calculation. Likewise, energy at the target clock is reported only when the corresponding routed netlist passes setup. This produces an explicit unavailable value for BCH rather than a favorable-looking estimate at an operating point the implementation cannot sustain.

No seed is discarded for an unfavorable metric, and no architecture receives a seed-specific density, floorplan, clock, or synthesis change. The protocol distinguishes an infrastructure failure from a design outcome: only the former could be retried, using the same seed and with the failed attempt preserved. The completed matrix needed no such replacements. Hsiao qualification is also prospective; its exact proof, structural check, source hash, and synthesis-policy check precede its physical measurements. These controls make the route set closed under the declared decision rules and prevent a later result from changing which measurements are admitted.

The unit of analysis depends on the claim. A physical point is one frozen hardware identity at one seed. A paired effect is the difference between the two SECDED points sharing that seed. An architecture summary aggregates five points only after the paired values are preserved. A correctness statement joins proof evidence to a hardware identity but not to a place-and-route seed. Keeping these units distinct prevents a mean PPA value from being mistaken for a proof, a formal proof from implying timing closure, or five deterministic routes from being described as statistical replication of fabricated chips.

IV. EXPERIMENTAL SETUP

All implementations protect a 64-bit payload. SECDED and Hsiao use 72-bit codewords; BCH uses 78. The container-pinned ORFS/OpenROAD flow targets SKY130HD at TT, 1.80 V, 25°C. Common settings are a 10 ns clock, 10% input/output delays, 0.05 pF output load, 35% core utilization, 0.55 placement density, and 1 worker. RTL, SDC, libraries, LEF/RC collateral, Docker digest, ORFS/OpenROAD revisions, trace, transaction contract, and power procedure are frozen.

Before execution, the seed rule was fixed as the first 5 primes at least 11: {11, 13, 17, 19, 23}. For seed s , the global-placement, global-routing, and general OpenROAD seed controls all equal s for every architecture. All 20 routes are fresh Revision-2 runs; no historical seed-11 directory is reused, no result is replaced, and the Hsiao RTL is frozen before its first route. The sweep is a controlled sensitivity test of placement/routing heuristics, not a random sample from an OpenROAD population.

For each route we preserve stage status, raw reports, run metadata, source/configuration hashes, and a SHA-256 artifact inventory. Extracted metrics include final standard-cell area and counts; setup/hold slack, TNS, violations, and Eq. (1); global/detailed wirelength, vias, congestion, and DRC; and, where eligible, internal, switching, leakage, total power, and energy/op.

The matrix is interleaved by seed: combinational SECDED, pipelined SECDED, algorithmic Hsiao, and BCH are run at one seed before advancing to the next. Each route must complete the common flow and preserve its final report even when timing fails. All 20 runs are DRC-clean and hold-clean; setup feasibility alone gates target-clock energy. The protocol makes SECDED power mandatory, Hsiao power conditional on meeting 10 ns, and BCH power unnecessary because its role is physical-feasibility contrast.

Power annotation uses the final ODB, SDC, and SPEF with the same interface VCD replayed across seeds. Activity is propagated into implementation-specific internal nets before a full-precision 12-digit power report. The comparison therefore retains clock-tree, buffering, placement, and routing consequences of each RTL. It does not isolate a causal energy contribution from registers or XOR depth, and it does not extrapolate voltage/frequency scaling.

V. RESULTS

A. Exact-Equivalent SECDED Robustness

The routed SECDED contrast is stable enough to remain architectural rather than seed-specific. Pipelining changes

TABLE II
FRESH FIVE-SEED POST-ROUTE SENSITIVITY STATISTICS (MEAN [MIN-MAX])

Architecture	Area (μm^2)	Slack-derived freq. (MHz)	Detailed wire (μm)	Energy (pJ/op)	10 ns
SECEDED comb.	18650 [18594–18728]	238.9 [225.6–245.1]	44917 [44609–45257]	145.77 [144.83–146.49]	5/5
SECEDED pipe.	25588 [25487–25640]	350.0 [347.5–355.8]	49964 [49703–50101]	111.63 [110.76–112.22]	5/5
Hsiao algorithmic	19191 [19136–19257]	212.9 [207.0–215.6]	49810 [49599–50142]	149.57 [148.87–150.77]	5/5
BCH (78,64,t=2)	57240 [56992–57763]	66.6 [65.9–67.7]	185035 [179964–191875]	–	0/5

standard-cell area by 37.2% on average, with a 36.7–37.8% matched-seed envelope. Detailed wirelength changes by 11.2% (10.6–12.1%), and the slack-derived frequency estimate changes by 46.6% (42.1–55.3%). The pipe is larger in 5/5 seeds, more routing-intensive in 5/5, and faster by Eq. (1) in 5/5.

The equal-work energy change is -23.4% on average (from -24.0% to -22.5%), with lower pipeline energy in 5/5 seeds. This is a measured result for the evaluated microarchitecture at a sustained common workload, not a general claim that pipeline registers save energy. It coexists with a seed-independent request-latency increase from one to three cycles and II=1 for both designs.

Variability is metric-dependent. The matched area effect has sample standard deviation 0.43 percentage points and only a 36.7–37.8% envelope; the wire effect has 0.58 points of spread. Energy is similarly narrow at 0.58 points. By contrast, the frequency effect has 5.07 points of spread and spans 42.1–55.3%. Thus its direction is robust in this sweep, while its magnitude is more sensitive to the placement/routing trajectory than area or energy. This is why all paired values are retained rather than replacing seed 11 with one aggregate.

The raw mean points give the architectural scale. Combinational SECEDED uses 18650 μm^2 , 2075 cells, and 44917 μm of detailed wire; pipelined SECEDED uses 25588 μm^2 , 2662 cells, and 49964 μm . Mean slack-derived frequency changes from 238.9 to 350.0 MHz, while energy changes from 145.77 to 111.63 pJ/op. Both have zero setup violations at every evaluated seed. These are means of five fresh routes, not historical seed-11 measurements reused as samples.

Exact per-seed enumeration finds the pair non-dominated in 5/5 seeds. At aggregate means the relation is non-dominated; the range-sensitive result is also non-dominated. The physical trade therefore cannot be reduced to one “better SECEDED” without a system preference over area, latency, frequency, and streaming energy.

Table III exposes the exact matched effects that underlie the envelope and summary markers. Cell-count change is especially stable because the five routes map almost the same logical structure, whereas detailed wire and slack retain a larger placement/routing component. The maximum frequency benefit does not occur at the seed with the maximum area increase, so the seed cannot be modeled as a single favorable-to-unfavorable scale shared by every metric. Area and energy changes also drift in opposite directions over parts of the sweep. These observations motivate pairing each metric independently instead of selecting one apparently representative route.

The decision conclusion is more stable than any single

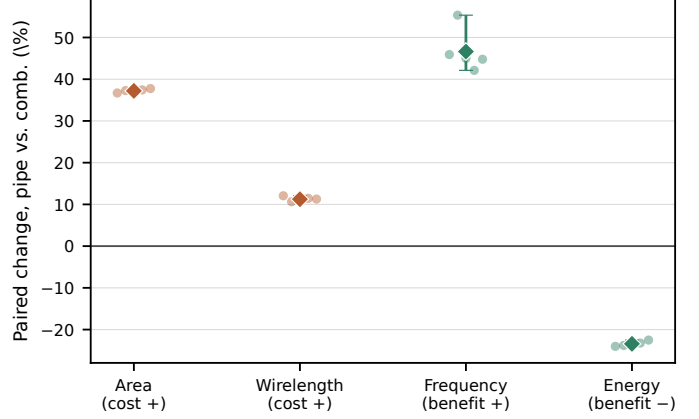


Fig. 2. Matched SECEDED effects. Light markers are the five seeds; diamonds and bars are mean and min-max. Positive area/wire values are costs, positive frequency is benefit, and negative energy is benefit.

magnitude. Every row keeps the same signs for area, wire, frequency, and energy, yet every row is non-dominated once request latency and the admitted objectives are included. The aggregate mean and conservative range test reach the same relation. Thus the five-seed result strengthens a directional statement—the evaluated pipeline consistently exchanges area and latency for frequency and streaming energy—without upgrading it into a population law. The remaining uncertainty concerns the size and portability of those effects, not whether one of the two frozen SECEDED identities wins all admitted objectives in this campaign.

B. Same-Code Hsiao Recovery

The algorithmic Hsiao identity routes in five fresh runs and meets 10 ns in 5/5 seeds. Its mean final standard-cell area is 19191 μm^2 (19136–19257), mean slack-derived frequency is 212.9 MHz (207.0–215.6), and mean detailed wirelength is 49810 μm (49599–50142). Its feasible-point energy statistic is reported in Table II; an unavailable value is retained as a dash rather than inferred.

Relative to combinational SECEDED at aggregate means, algorithmic Hsiao changes area by 2.9%, cell count by 4.8%, detailed wirelength by 10.9%, slack-derived frequency by -10.9%, and energy/op by 2.6%. These differences do not isolate a “Hsiao code penalty”: the parity-check matrix, encoder network, syndrome logic, correction matching, and routed realization differ together. Instead, Hsiao adds a third measured implementation in the W1-correct/W2-detect guarantee tier.

TABLE III
EXACT MATCHED SECEDED EFFECTS BY SEED (PIPELINED RELATIVE TO COMBINATIONAL, %)

Seed	Area	Cell count	Detailed wire	f_{slack}	Energy/op	Pareto relation
11	36.7	28.3	12.1	45.9	-24.0	non-dominated
13	37.3	28.2	10.6	55.3	-23.8	non-dominated
17	36.8	28.1	10.8	45.0	-23.5	non-dominated
19	37.4	28.3	11.5	42.1	-23.2	non-dominated
23	37.8	28.4	11.3	44.8	-22.5	non-dominated

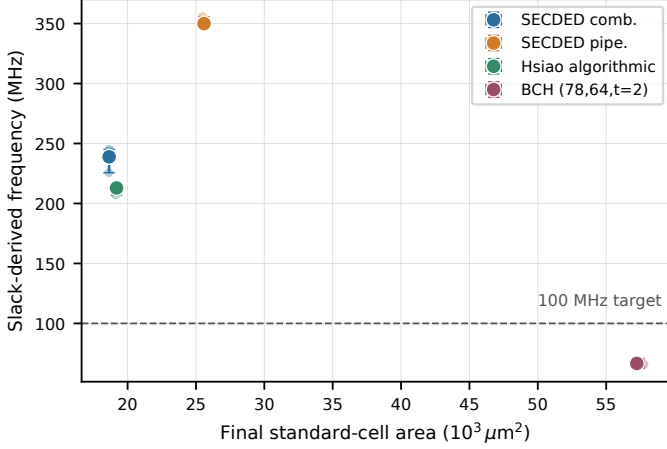


Fig. 3. Five fresh seed points (light), means (large markers), and min-max bars. Frequency is the fixed-implementation signed-slack estimate under the 10 ns optimization, not a sweep.

The result also distinguishes policy failure from physical inferiority. The historical table RTL stopped before mapping because inferred storage exceeded the unchanged synthesis-memory limit; it never established an area or timing point. The algorithmic RTL avoids that representation boundary while keeping the same limit, common floorplan policy, and code behavior. Its measured outcome is prospective evidence for a new hardware identity, not a selectively tuned historical run.

This point does not retroactively fill the old table-based hardware row. The exact proof connects both RTLs to the same behavioral/code semantics, but the measured PPA belongs to the prospectively frozen comparison-logic decoder. That distinction is precisely the identity discipline the paper advocates.

C. Stronger-Correcting BCH Feasibility

The evaluated BCH syndrome/Chien realization meets the common 10 ns target in 0/5 seeds. Across the sweep it has mean area $57240 \mu\text{m}^2$ (56992–57763), slack-derived frequency 66.6 MHz (65.9–67.7), and detailed wirelength $185035 \mu\text{m}$ (179964–191875). Its stronger W2-correction guarantee remains valuable, but physical conclusions apply only to this RTL and common boundary. Recent BCH architectures demonstrate that other realizations can occupy different cost/latency regions [7]; the repository contained no second correction-qualified (78, 64, $t = 2$) architecture eligible for this campaign.

The timing miss is consistent rather than marginal: signed setup slack averages -5.01 ns and ranges from -5.17 to -4.78 ns,

with 66 setup violations at every seed. Relative to combinational SECEDED means, BCH changes area by 206.9%, cell count by 243.7%, detailed wirelength by 311.9%, and slack-derived frequency by -72.1%. All routes are DRC- and hold-clean, so this is a completed physical point that fails the common setup target, not a route crash. No 100 MHz energy is displayed because that clock is infeasible for all five runs.

D. Cross-Tier Design-Space Reading

Within the shared SECEDED guarantee tier, combinational SECEDED, pipelined SECEDED, and algorithmic Hsiao offer different physical choices. The exact-equivalent SECEDED pair is non-dominated in every seed because the combinational design retains area and latency advantages while the pipeline retains frequency and energy advantages. Hsiao supplies the same categorical guarantee through a different code and hardware identity; Table II makes its measured area, timing, routing, and energy visible without claiming exact semantic equivalence to extended Hamming.

BCH occupies a stronger guarantee stratum. If a system requires correction of every two-coordinate error, none of the SECEDED points can replace it merely because they meet timing. Conversely, under the evaluated 100 MHz feasibility requirement, this BCH realization is not an achievable candidate. Exact enumeration therefore reports conditional choices rather than assigning a numeric value to correction strength. This preserves both facts: the measured SECEDED realizations are physically favorable under the stated objectives, and the measured BCH realization guarantees correction that they do not.

VI. DISCUSSION AND LIMITATIONS

The SECEDED experiment shows why code semantics, architecture, and physical outcome must be separately named. Exact equivalence permits causal attribution to the evaluated temporal microarchitecture as a whole, while matched seeds test whether its trade direction is an artifact of one heuristic trajectory. The envelope is intentionally descriptive: five fixed seeds cannot estimate a universal distribution, tail probability, confidence interval, or tool-independent effect.

Hsiao recovery supplies a second lesson. The old decoder's policy failure concerned representation, not code quality. Replacing its inferred table with explicit comparisons preserved behavior and passed the unchanged flow, but created a new hardware identity whose PPA cannot be assigned to the table implementation. Conversely, exact equivalence is what permits

the prior guarantee semantics to transfer without repeating an ambiguous code-level experiment.

Correction tiers must remain separate. Within the stated physical objectives, an SECDED point may be Pareto-superior to the evaluated BCH realization, yet it cannot correct every double-coordinate error. A system with mandatory W2 correction may accept a timing gap or seek another BCH/DEC architecture. The data therefore support conditional selection rather than a family-wide dominance statement. Weight-3 counts are omitted from the main results because their unweighted canonical masks have no demonstrated operational probability; they remain useful artifact-level decoder checks.

The quantitative scope is one open 130-nm library, one TT 1.80 V/25°C corner, one 10 ns target, one floorplan policy, and specific RTL. OpenSTA activity power is an estimate rather than silicon measurement. We exclude SRAM macro/storage organization, PVT variation, IR drop, clock-domain integration, scrubbing, interleaving, physical bit adjacency, FIT/SER, and field-weighted SDC/DUE. Common policy reduces configuration confounding but does not prove optimal floorplanning for each architecture. These limits bound numerical generalization while leaving the within-flow identity and sensitivity results independently auditable.

A. Selection Implications

The evidence suggests a staged selection rule. First filter implementations by categorical guarantee, target feasibility, maximum request latency, and evidence status. Only then compare surviving hardware identities on area, routing, and achievable energy. A one-cycle latency requirement favors combinational SECDED over its exact-equivalent pipeline even though the latter has higher frequency and lower streaming energy. A strict W2-correction requirement admits the BCH semantics but exposes a physical implementation gap at 10 ns. Hsiao is now a measured same-guarantee alternative rather than an unavailable label, yet its aggregate outcome cannot be assigned to all Hsiao architectures.

This ordering also clarifies what future work could change. A multi-corner sweep could test whether timing and power orderings survive PVT variation; another library or node could change XOR/register cost ratios; a second qualified BCH decoder could show whether the present target miss is architectural; and a physical memory-fault model could weight beyond-guarantee masks. Each extension should create a new named physical or hardware identity rather than overwrite this dataset. None would alter the exact equivalence proof or the measurements of the frozen implementations.

B. Claim Boundaries and Portability

The evidence separates conclusions with different portability. Exact RTL equivalence is independent of the route seeds and establishes behavior for the frozen SECDED pair and Hsiao decoder relation. Exhaustive guarantee tests establish categorical behavior only for the qualified code/decoder universe. Timing, area, routing, and activity-power results are properties of the frozen hardware and physical identities; their numerical values

should not be transferred to a different decoder, constraint set, library, corner, or tool revision. Finally, seed-ordering statements are properties of this closed five-seed sensitivity set. Making these levels explicit avoids both underclaiming the proofs and overclaiming the physical data.

Several tempting inferences are therefore excluded. The frequency metric does not say the designs were reimplemented at their maximum clock. Equal-work no-error energy does not predict energy under correction-heavy traffic. A DRC-clean BCH route does not imply target feasibility, and a target miss does not invalidate the BCH correction guarantee. Hsiao's successful algorithmic implementation does not establish that all Hsiao matrices or syndrome organizations have similar PPA. Likewise, stable ordering over five matched seeds is evidence against a one-seed artifact under the pinned heuristic controls, not evidence that another flow, node, or seed range must preserve it.

These boundaries still allow useful engineering portability. The identity/evidence method, missing-data rules, matched-pair construction, and categorical treatment of correction strength can be reused without assuming the present numerical effects. A replication can substitute another library, corner, or qualified implementation while retaining the same joins and decision logic, then compare whether signs, feasibility, and Pareto relations change. Reporting a divergence would be informative rather than contradictory because it would attach the new outcome to a newly named physical identity.

C. Auditability

The artifact retains full-precision records even though the paper rounds for display. Each route has a unique run ID, complete stage status, raw ORFS JSON and reports, metadata, and SHA-256 inventory. Post-execution analysis verifies every raw manifest before generating the per-seed CSV, architecture summaries, paired-effect JSON, and manuscript registry. Figures and tables are regenerated from that registry; an independent validator recomputes headline macros, rejects ambiguous timing terminology, and rechecks frozen Hsiao and Revision-1 hashes. This discipline makes each cross-layer claim independently auditable without making provenance itself the scientific contribution.

VII. CONCLUSION

ECC code labels do not uniquely determine hardware cost. By qualifying identity before PPA and applying a fresh five-seed common-policy sweep, we turn a single-seed observation into a matched sensitivity result: the exact-equivalent SECDED pair preserves distinct area/latency/frequency/energy trade-offs across the evaluated seeds. Exact proof also enables a same-code algorithmic Hsiao hardware identity to enter the common flow without relaxing synthesis policy, while the BCH result remains an explicitly scoped stronger-correction feasibility point. The appropriate design decision is therefore conditional on guarantee, target feasibility, and implementation identity—not a family label or a scalar score.

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